

Nguyen Thanh Long

Full-Stack Engineer

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SUMMARY

Full-stack engineer, 2.5 years, working in TypeScript on both ends of a production IoT platform: the Next.js dashboard partners use and the NestJS APIs behind it. I am currently the only engineer on the cloud side, so the same work also covers the database, the message pipeline and the deployments.

EXPERIENCE

Full-Stack Engineer at Rogo Solutions JSC Jan 2024 to present · Hanoi

A 20-person IoT startup. I am the only engineer on the cloud side, so the dashboard, the API, the message pipeline and the deploy path are all mine. The platform serves 640k users and 69k devices for 26 partner projects, at roughly 2.6M API requests a day and 15.5k concurrent MQTT connections.

- Permissions are **attribute based**. A partner's rights are stored as resource and action pairs scoped to a project, then encoded into a short string so the whole set fits inside a **Firestore custom claim** and travels with the ID token, which auth middleware decodes per request. The dashboard builds **CASL** abilities from the same data, so the interface only offers actions the API would accept. I wrote v2 of this and the migration off v1.
- Device state moves over **MQTT into Kafka** and out to MongoDB. The consumer buffers and **coalesces** repeated updates rather than writing per message, and commits its offset after the write, so losing a pod costs latency rather than data. **KEDA** scales it between 2 and 12 replicas on **consumer group lag**, with a floor of two so a node drain does not leave the topic unconsumed. Partner traffic used to run through a single unreplicated broker.
- There was no monitoring when I started. **Prometheus, Loki and Grafana** now cover the three clusters with the AWS side pulled in from **CloudWatch**, so an incident starts at a dashboard instead of a guess. I also own the staging to production path across four repositories.
- The test suite had been abandoned because running it exhausted memory and froze dev machines, so nothing was gated on it. Moving to **Jest 29** with transpile only tests and a separate type check brought it back to a few seconds, and back into CI. The dashboard also survives a CMS outage now instead of crashing.

SELECTED PROJECTS

Educrysal (Next.js, Supabase, Tailwind, PWA) · educrysal.vercel.app
A guided crystal-growing app for children: step-by-step programs, a private photo diary, and a gallery a moderator approves. Access rules live in Postgres row level security rather than in the application, so a route cannot leak another child's diary. Installable as a PWA. Six months of work.

IoT Cloud MCP (TypeScript, NestJS, MCP SDK, Redis)
An MCP server for IoT cloud platforms, so an assistant can query and control devices through one interface rather than a separate integration per platform.

Novel Studio (NestJS, PostgreSQL with pgvector)
A long-form writing tool. The awkward part is that prompt cost grows with the story, so it keeps a canon store, a rolling summary and vector retrieval instead of resending history.

TECHNICAL SKILLS

Frontend	TypeScript, React, Next.js, Tailwind CSS, Ant Design, React Query, React Native (Expo)
Backend	Node.js, NestJS, Express, REST APIs, WebSockets
Data	PostgreSQL, MongoDB, Redis, MySQL
Infrastructure	Docker, Kubernetes, AWS (ECS, S3, Lambda), Terraform, GitHub Actions
Messaging & auth	Kafka, MQTT, Firebase Auth, Keycloak, JWT

EDUCATION & ADDITIONAL

B.Sc. in Software Engineering, FPT University 2022 to 2026

Final-year project: a real-time multiplayer game backend (ASP.NET Core, WebSockets, MySQL) using lag compensation to keep play fair across uneven network conditions.

Certifications: Project Management Principles (UC Irvine, 2025) · Software Development Lifecycle Specialization (University of Minnesota, 2024).

Coach competitive-programming students for national Informatics examinations, weekly since 2024. Freelance backend work alongside full-time employment. English: professional working proficiency.